

ASHUTOSH SHUKLA

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Education

Indian Institute of Information Technology, Gwalior

(Expected) 06/2027

Integrated Bachelors + Masters in Information Technology

CGPA : 8.92/10

Relevant Coursework: AI/ML, Natural Language Processing, Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, OOPs

Experience

Advance Engineering Intern

05/2026 – 07/2026

Accenture

Pune, India

- Developed an AI enterprise assistant over a 7-phase implementation cycle using **FastAPI** and **Gemini API**, integrating a **RAG-based** knowledge base to dynamically route natural-language queries to verified endpoints.
- Authenticated the chatbot access via **JWT**, and integrated a dual-database architecture leveraging **Redis** for active context caching and **PostgreSQL** for long-term chat history retention.
- Automated **AWS S3** data lifecycle management using **aws-sdk-go-v2** library and a concurrent **Golang** multi-format compression engine (ZIP and TAR) that slashed manual archival time by **90%** (from 50 minutes to under 5 minutes).

Software Engineering Intern

12/2023 – 03/2024

Mitt Arv

Hyderabad, India; Remote

- Collaborated on the Home Page Tutorial for the mobile App, encompassing **7** key on-boarding steps with gesture detection and over **10** vital buttons and design patterns to improve engagement and promote landing screen utility.
- Enhanced user experience by resolving more than **12** routing and state management issues using **GetX** (for state) and **Navigator** (for routing) across key modules like Asset Vault, Emotional Will, and other critical sections.

Projects

Ting – Custom-built Search Engine | Python, FastAPI, MS Marco Dataset

03/2026 – 04/2026

- Teamed up to Engineer a full-stack semantic search engine over **8.8M** documents, supporting 8 lexical, dense, and hybrid retrieval strategies through an interactive interface built over **ElasticSearch**, evaluated by **NDCG** and **MRR**.
- Designed a 4-stage Learning-to-Rank (L2R) architecture leveraging **BM25** candidate generation, **15+** proximity features, **Sentence-Transformer** embeddings, and **XGBoost** ranking models for relevance optimization.
- Implemented 2 significant retrieval enhancement techniques — **MMR** reranking and **TopRM3** query expansion to improve result diversity, relevance, and query understanding across multiple search modes.

Komms – True P2P Chat & File Sharing App | Python, WebSockets, Multithreading, PyQt6

01/2025 – 04/2025

- Crafted a decentralized P2P application for real-time communication and secure file exchange, leveraging **socket programming** and **RSA** public-key encryption system.
- Built an asynchronous file transfer pipeline featuring **1MB** chunking, resumable transfers, and **SHA-256** verification, while improving peer discovery and concurrency using **UDP** broadcasts, **WSS**, and **QThreadPool**.

Technical Skills

Languages: C, C++, Dart, Golang, Java, Python, SQL

Frameworks & Libraries: Flutter, Firebase, PyTorch, Scikit-Learn, XGBoost, CatBoost, Pandas, NumPy, Matplotlib

Search & Data Systems: Elasticsearch, FAISS, PostgreSQL, MySQL

Developer Tools: Docker, Git, GitHub, Linux, FastAPI, REST APIs

Achievements

- Achieved an accuracy of **91.56%** in **Flipkart GridLock** - A Traffic Modelling Hackathon.
- Secured **3rd** position out of 100+ participants in **WJPS** - a coding contest organized by IIIT Gwalior's Tech Forum.
- Awarded **1st** Prize in the Development Track out of 150+ competitors in **AASF** - Winter Projects.

Position of Responsibility

Events and Planning Head

Student Gyan Movement (Student Activity Council - Welfare Department) | Homepage

- Educated **300+** underprivileged students in the nearby areas of Gwalior and conducted over **20** school drives and value-inculcating events for the all-round development and mental well-being of the children.